

Non-Executing Security Analysis of AI Model Supply-Chain Artifacts

Provenance, Serialization, Impersonation, and Repository Risk Inspection

github.com/poojakira/nf-model-provenance-scanner - public, default branch main.

EVIDENCE POSTURE

Verified, scoped, reproducible

Claims are constrained to the repository evidence snapshot and stated limitations.

241

Tests passed

1

Tests skipped

6

Additional pytest subtests passed

75.81%

Statement coverage

SYSTEM VIEW

Architecture and Threat-Control Flow

Trust Boundary

Input surface

Data, models, requests, policies, or artifacts entering the tested path.

Control surface

Repository-specific validation, analysis, and fail-closed checks.

Evidence surface

Findings, metrics, tests, logs, or reproducible artifacts.

This diagram describes the tested repository path, not a universal deployment guarantee.

01

Enumerate repository and model artifacts.

02

Parse supported serialization/model formats.

03

Apply AST, pattern, symbolic, taint-style, metadata, and provenance checks.

04

Reject unsafe dynamic-execution paths in the scanner itself.

05

Emit findings and fail-loud states for incomplete or unsafe analysis.

Research Contribution

A primarily non-executing model-artifact security scanner combining static code analysis, symbolic and taint-style inspection, binary-format parsing, provenance checks, and fail-loud handling of incomplete analysis before model loading.

Verified Evidence

EV-01

CI coverage gate: 75%

Method and Control Path

01

Enumerate repository and model artifacts.

02

Parse supported serialization/model formats.

03

Apply AST, pattern, symbolic, taint-style, metadata, and provenance checks.

04

Reject unsafe dynamic-execution paths in the scanner itself.

05

Emit findings and fail-loud states for incomplete or unsafe analysis.

Limitations and Residual Risk

- Static analysis cannot prove a model is safe.
- No claim is made that neural weight backdoors are generally detected.
- Fixture-level results do not establish population-level precision or recall.
- Unsupported or incomplete parsing must remain fail-loud rather than silently clean.

Reproducibility

Expected evidence at the cited snapshot: 241 passed, 1 skipped, 6 subtests passed, 75.81% coverage.

Evidence Provenance

Claim discipline

Metrics come from the repository's verified metrics and research brief.

Scope discipline

Benchmarks are reported only at their documented fixture, dataset, or test scope.

Reproduction

PDF and PNG are generated locally from committed evidence without paid services.